

Industrial Doctor - VibroKit

Up to 4 3-axial accelerometer vibration measurement solution



Features

JNK-01

- Dynamic Range: $\pm 16g$
- Resolution: 2048 LSB per g
- Modbus-RTU over RS-485 interface
 - Support Modbus multi-read/write (FUNC 03/06/16) command
 - RS-485 baud rate up to 4MHz
- Up to 12.8 kHz ODR, 6.3 kHz bandwidth with programmable digital low-pass filters
- Single +5V operation voltage
- -40 °C to +85 °C operation temperature
- Power Consumption
 - Operation: typical 30 mA
 - Standby: 20 mA
- 5000 g shock tolerance
- RoHS-compliance
 - IP67 rating when cable connected
 - 5m cable with Min-DIN-6 connector
 - $\phi 27 \times H13.3 \text{ mm} / \phi 5.5 \text{ mm}$ screw hole



Descriptions

The JNK-01 is a compact and robust vibration sensor equipped with high performance MEMS accelerometer. It features a RS-485 digital interface supporting Modbus-RTU protocol that can be easily integrated into existing industry infrastructure. A companion hub JHN-01 is a 4-Port Modbus gateway that can bridge up to 4 JNK-01 sensor nodes to Ethernet. The bridging is accomplished by conversion between Modbus-RTU and Modbus-TCP communication protocols.

JHN-01

- Bridge between Modbus-RTU and -TCP protocols
- 4 Serial ports:
 - RS-485 interface
 - Modbus-RTU Master
- 1 Ethernet ports:
 - 10/100BaseTX Ethernet
 - 1 Modbus-TCP Server (Slave) for gateway configuration
 - 4 Modbus-TCP Server (Slave) for sensor node bridging
 - 4 TCP Server for sensor data stream
- Single +24V operation voltage
- 4x +5V power out to JNK-01 nodes
- -40°C to +85°C operation temperature
- Configuration options: web console and Modbus-TCP Server (Slave)
- Embedded NVM for data logging
- RoHS-compliance aluminum case:
 - 4x female Min-Din-6 for serial port and +5V power out to Jupiter nodes
 - 1x GPIOCONN w/ 8-Input/8-Output
 - 1x RJ45 for Ethernet connection
 - 1x PWRCONN for +24V power in
 - 1x RESET push button
 - 1x Power LED
 - 150 x 110 x 35 mm



GPIOCONN

- 8-Input
- 8-Output

RS485-3

RS485-1

RS485-4

RS485-2



Reset button

RJ45 for Ethernet connection

PWRCONN for +24V power in

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Specifications

JNK-01

Environment temperature $T_a = 25^{\circ}\text{C}$ if not specified otherwise

Parameter	Conditions	Min.	Typ.	Max.	Unit
Operation voltage (VDD5V)	$T_a = -40^{\circ}\text{C} \sim +85^{\circ}\text{C}$	3.8	5.0	5.5	V
Operating current		20	30	40	mA
Suspend current		—	20	—	mA
Operation temperature		-40	—	+85	$^{\circ}\text{C}$
Storage temperature		-40	—	+125	$^{\circ}\text{C}$
Accelerometer					
Dynamic range		—	± 2 ± 4 ± 8 ± 16	—	g
Sensitivity		—	16384 8192 4096 <u>2048</u>	—	LSB per g
ODR (Output Data Rate)		—	25 <u>50</u> 100 200 400 800 1600 3200 6400 12800 25600	—	Hz
ODR accuracy	INTRS (CCh[0]) = 1'b0 compatible sampling	—	± 30	—	ppm
ODR accuracy	INTRS (CCh[0]) = 1'b1 intrinsic sampling	+3*	+4*	+6*	%
Bandwidth		—	6300	—	Hz
Sensitivity to temp. dependency	$T_a = -40^{\circ}\text{C} \sim +85^{\circ}\text{C}$, deviation from $T = +25^{\circ}\text{C}$	—	± 1	± 2	%
Noise density	X/Y-axis Z-axis	—	110 160	—	$\mu\text{g}/\sqrt{\text{Hz}}$

JHN-01

Environment temperature $T_a = 25^{\circ}\text{C}$ if not specified otherwise

Parameter	Conditions	Min.	Typ.	Max.	Unit
Operation voltage (VDD24V)	T _a = -40°C ~ +85°C	—	24.0	—	V
Operating current	Gateway itself only, excluding sensor power	—	0.5	—	A
NVM for data logging		—	27	—	GB
Operation temperature		-40	—	+85	°C
Storage temperature		-40	—	+125	°C
Ethernet Interface					
10/100Base TX Port	×1				
Magnetic isolation protection	1.5kV				
Connector	RJ45				
Ethernet Services					
Service Name	Port Number	Description			
MBAP, Gateway	502	Gateway Modbus-TCP Server (Slave)			
MBAP, SN1	505	Sensor node SN1 Modbus-TCP Server (Slave)			
MBAP, SN2	506	Sensor node SN2 Modbus-TCP Server (Slave)			
MBAP, SN3	507	Sensor node SN3 Modbus-TCP Server (Slave)			
MBAP, SN4	508	Sensor node SN4 Modbus-TCP Server (Slave)			
SN1 Data Stream	1505	Sensor node SN1 data stream TCP Server			
SN2 Data Stream	1506	Sensor node SN2 data stream TCP Server			
SN3 Data Stream	1507	Sensor node SN3 data stream TCP Server			
SN4 Data Stream	1508	Sensor node SN4 data stream TCP Server			
Serial Interface					
No. of Ports	4				
Connector	Female Min-Din-6, with +5V power out				
Baud rate	Up to 4MHz				
Data bits	8				
Parity	None				
Stop bits	1				
Flow control	None				
RS-485-2w	Data+, Data-, GND				
Terminator for RS-485	120 ohms				
Industry protocol	Modbus-RTU Master				
LED					
R-LED-P	Red power LED. Light-up when VDD24V connect to +24V				
G-LED-S	Green status LED. Not used.				
G-LED-1~4	Green sensor node status LEDs. Not used.				
R-LED-1~4	Red sensor node status LEDs. Not used				
Button					
RESET	Reset push button. Push RESET to reset JHN-01.				

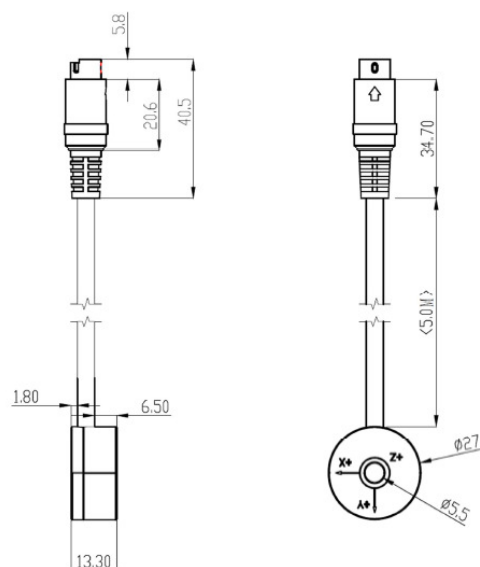
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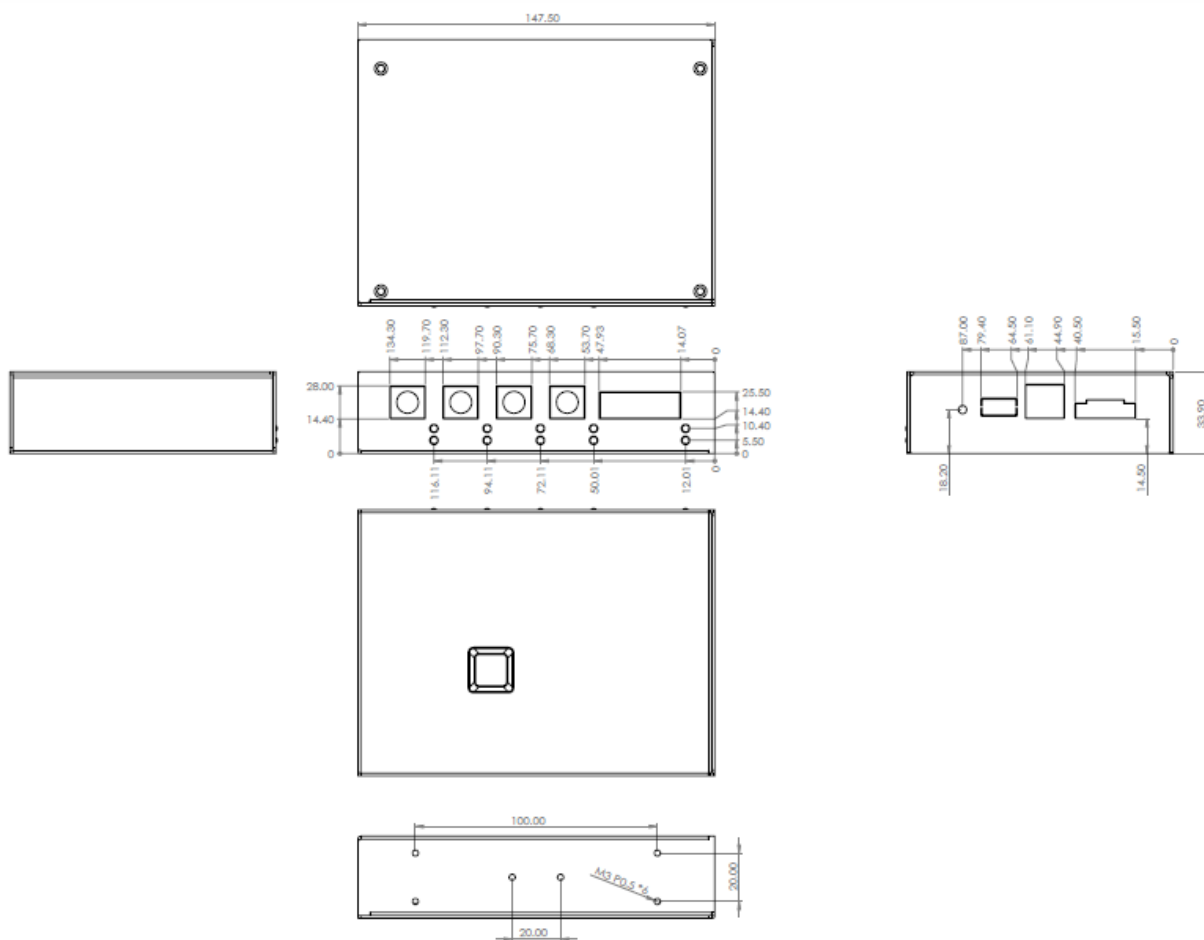
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Dimensions

JNK-01



JHN-01



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